QIntern 2026 Project Proposal

Mixed-State g-Purity: Purification Consistency, Monotonicity, and Trade-offs

Mentor: Oxana Shaya | Duration: 1 July–15 August 2026

Project summary

Recent group-Fourier-decomposition (GFD) methods characterize pure-state resourcefulness by decomposing a state into group-representation sectors and measuring the squared weight in each sector. This project asks how such g-purity profiles should be extended to mixed states. The central point is that there is no unique "obvious" extension: one may decompose ρ, decompose $\sqrt{\rho}$ through the canonical purification, or use a convex-roof/resource-monotone construction. The project will map the trade-offs among these choices, with special attention to purification consistency and monotonicity under free operations.

Motivation

Mixed-state diagnostics are needed for noisy experimental states, thermal states, and convex mixtures. A useful mixed-state g-purity should retain the representation-theoretic information of the pure-state GFD profile while behaving sensibly as a resource-theoretic quantity. In particular, the project will not assume in advance that one definition is canonical. Instead, it will compare natural candidates according to explicit criteria: pure-state compatibility, purification consistency, monotonicity/data processing under free operations, convexity, and computability.

Candidate profiles

Let a compact group G act on $\mathcal{H}$ by U_g, and let the adjoint action $X \mapsto U_g X U_g^\dagger$ decompose $\mathcal{B}(\mathcal{H})$ into isotypic components with orthogonal projectors Π_λ.

Operator-space profile. The most direct extension decomposes the density matrix itself:

$$p_\lambda^{(1)}(\rho) = \frac{\|\Pi_\lambda(\rho)\|_2^2}{\operatorname{tr}(\rho^2)}.$$

This is natural from the Liouville-space point of view and is technically friendly, but its normalization gives a *relative* sector weight inside ρ.

Purification-consistent profile. The canonical purification $|\sqrt{\rho}\rangle\rangle \in \mathcal{H} \otimes \mathcal{H}_R$ transforms under $U_g \otimes \overline{U_g}$ exactly as $\sqrt{\rho}$ transforms by conjugation. Hence a strong candidate is

$$p_\lambda^{(1/2)}(\rho) = \|\Pi_\lambda(\sqrt{\rho})\|_2^2, \qquad \|\sqrt{\rho}\|_2^2 = \operatorname{tr}\rho = 1.$$

Thus $p^{(1/2)}$ is precisely the GFD profile of the canonical purification under the doubled action. This is the cleanest motivation for privileging $\alpha = 1/2$, but it does not automatically imply resource monotonicity.

Power family and convex-roof benchmark. A unifying family is

$$p_\lambda^{(\alpha)}(\rho) = \frac{\|\Pi_\lambda(\rho^\alpha)\|_2^2}{\|\rho^\alpha\|_2^2}, \qquad \alpha > 0,$$

with $\alpha = 1$ and $\alpha = 1/2$ as the main cases. A further benchmark is a convex-roof extension of the pure-state quantity,

$$p_\lambda^{\mathrm{roof}}(\rho) = \inf_{\rho = \sum_i q_i |\psi_i\rangle\langle\psi_i|} \sum_i q_i p_\lambda(|\psi_i\rangle),$$

which is more resource-theoretic but usually harder to compute. The internship will treat the convex roof mainly as a conceptual benchmark, not as the primary implementation target.

Technical scope notes

For pure $\rho = |\psi\rangle\langle\psi|$, all ρ^α coincide with ρ. However, the doubled-vector profile of $|\rho\rangle\rangle \cong |\psi\rangle \otimes |\overline{\psi}\rangle$ need not have the same irrep labels as the original ket-level GFD profile of $|\psi\rangle$; the project will state this relation explicitly. The comparison will also distinguish absolute weights, relative weights, and twirl-subtracted diagnostics based on $\rho - \Phi_G(\rho)$, where Φ_G is the group twirl. The latter is useful for asymmetry-type resources because it isolates the non-invariant part of the state.

Objectives

- Formalize $p^{(1)}$, $p^{(1/2)}$, $p^{(\alpha)}$, and the convex-roof benchmark, including normalization conventions.
- Prove the purification identity: the $p^{(1/2)}$ profile equals the GFD profile of $|\sqrt{\rho}\rangle\rangle$ under $U_g \otimes \overline{U_g}$.
- Check basic properties: positivity, normalization, G-covariance invariance, pure-state compatibility, and behavior at maximally mixed states.
- Make monotonicity/data processing a primary test: determine which candidates are non-increasing, or fail to be non-increasing, under selected free or G-covariant operations.

- Work out explicit examples: qubit coherence/asymmetry, depolarized pure states, and Werner or isotropic two-qubit families.
- Produce a small reproducible Python/Jupyter prototype for the chosen finite-group or low-dimensional examples, rather than a fully general sector-projector package.

Work plan

Period	Main tasks	Milestone
1–7 July	Read core papers and selected background on resource monotones/asymmetry; choose two concrete example settings.	Scope and notation note.
8–14 July	Define the candidate profiles, normalizations, pure-state compatibility, and purification criterion.	Definition sheet.
15–22 July	Prove basic lemmas; analyze G-covariance invariance and first monotonicity/DPI checks.	Lemma and criteria draft.
23–31 July	Compute hand-checkable examples: qubit coherence/asymmetry, depolarized pure states, Werner/isotropic states.	Verified examples.
1–8 August	Implement the selected examples and generate noise/mixing plots; look for counterexamples to monotonicity.	Reproducible notebooks.
9–15 August	Final comparison table, report, and short presentation.	Final deliverables.

Deliverables

- A concise report comparing the ρ-, $\sqrt{\rho}$-, ρ^{α}-, and convex-roof viewpoints.
- A property table covering normalization, purification consistency, pure-state compatibility, convexity, monotonicity/DPI behavior, and computational cost.
- Worked examples with at least one explicit noise trajectory and one two-qubit family.
- Documented Python/Jupyter notebooks reproducing examples, plots, and numerical checks.
- A short final presentation for the QIntern closing session.

Student background

The project is suitable for a student with basic quantum information, linear algebra, and Python experience. Familiarity with group representations or quantum resource theories is helpful but not required.

Core references

1. P. Bermejo et al., *Characterizing quantum resourcefulness via group-Fourier decompositions*, arXiv:2506.19696 (2025).
2. N. L. Diaz et al., *A unified approach to quantum resource theories and a new class of free operations*, arXiv:2507.10851 (2025).
3. A. E. Deneris et al., *Analyzing the free states of one quantum resource theory as resource states of another*, arXiv:2507.11793 (2025).
4. I. Marvian and R. W. Spekkens, *Modes of asymmetry*, Phys. Rev. A 90, 062110 (2014).
5. E. Chitambar and G. Gour, *Quantum resource theories*, Rev. Mod. Phys. 91, 025001 (2019).